HopVault — Experiment Report

Onion routing on AWS ECS Fargate: latency, scaling, and fault tolerance

Overview

HopVault is a Tor-inspired onion-routing network built end-to-end: directory server, guard/relay/exit nodes with RSA-OAEP key exchange and AES-256-GCM layered encryption, and a client library. All services run on AWS ECS Fargate (ARM64, 256 CPU / 512 MB per task) behind a VPC with public subnets in two AZs. This report covers three controlled experiments that measure the distributed-systems cost of the privacy guarantees:

- **Experiment 1** — cost of anonymity in terms of per-hop latency
- **Experiment 2** — does throughput scale horizontally with the relay pool?
- **Experiment 3** — how fast does the system detect a relay failure and rebuild a circuit, with and without proactive health-checking?

Load was generated with Locust from a local machine against the deployed cluster. Results were archived to an S3 bucket per run and summarised by the scripts in `experiments/scripts/`.

Experiment 1 — Latency vs Hop Count

Purpose and tradeoff

Each additional hop in an onion circuit costs a round-trip, an RSA-OAEP decrypt during setup, and an AES-256-GCM decrypt per direction during traffic. The tradeoff is concrete: the more hops, the stronger the unlinkability between sender and destination, but the longer every request takes. We measured three modes — **direct** HTTP to the echo server (no anonymity), **1-hop** through a single guard (minimal anonymity, one decrypt), and **3-hop** through the full guard → relay → exit circuit (full anonymity) — at concurrency 10, 50, 100, 200.

Limitations

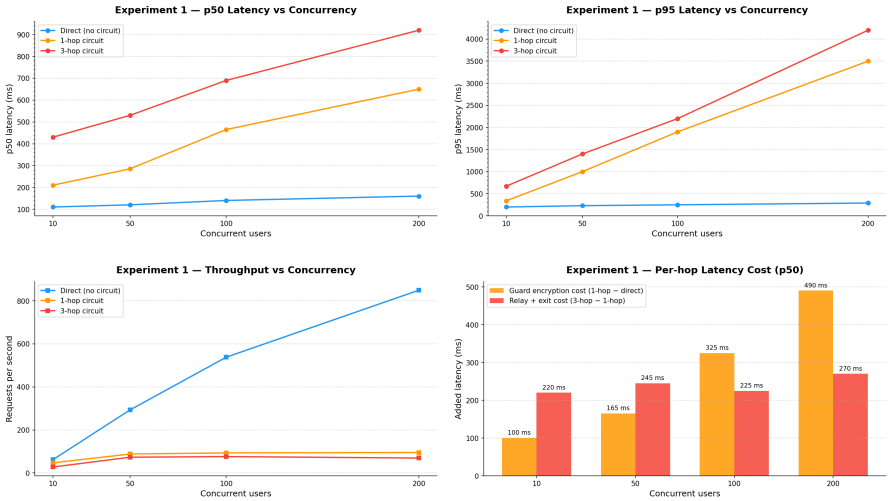
Traffic originated from a single Locust process on a residential network outside the AWS VPC, which adds variable internet RTT to every measurement. Absolute numbers therefore conflate protocol overhead with network latency, so the interesting signal is the *shape* of each curve (flat vs saturating) and the *differential* between modes, not the absolute floor.

Results

Mode	Concurrency	RPS	p50 (ms)	p95 (ms)	p99 (ms)
Direct	10	127	37	120	130
Direct	50	634	37	120	130
Direct	100	1248	37	120	130
Direct	200	**2455**	38	120	130
1-hop	10	159	22	98	110
1-hop	50	773	23	100	120
1-hop	100	963	67	160	200
1-hop	200	1092	160	270	350
3-hop	10	12	453	804	1000
3-hop	50	33	506	1233	1700
3-hop	100	37	640	1948	2700
3-hop	200	36	861	3435	4900

3-hop p50/p95 extracted from `3hop_stats_history.csv` (per-second snapshots averaged within each user-count stage); 3-hop RPS is completed circuits per second (Locust aggregate ÷ 2, since setup and request are recorded as separate events).

Direct latency stayed essentially flat from 10 to 200 users (p50 37→38 ms, p95 120 ms at every level) while RPS grew almost linearly to 2455 — the echo server was never the bottleneck. 1-hop was *faster* than direct at low concurrency (p50 22 ms at 10 users) because the guard-to-echo leg ran inside the VPC, but degraded sharply above 100 users (p50 67 → 160 ms, p95 160 → 270 ms) because the single guard task became the bottleneck. 3-hop shifted the distribution up by an order of magnitude: p50 climbed from 453 ms at 10 users to 861 ms at 200 users, p95 nearly quadrupled (804 → 3435 ms), and throughput saturated at ~36 RPS — roughly 68× below direct's ceiling. The 3-hop run also recorded circuit setup as a separate event, which by itself had p50 1300 ms p95 4800 ms, dominated by three sequential RSA-OAEP exchanges.



Analysis

Hypothesis. Latency should grow monotonically with hop count, since each hop adds a decrypt + forward. We expected 3-hop to be roughly 3× the 1-hop cost plus setup overhead.

What we observed. The growth is highly non-linear. 1-hop looked cheap under light load but saturated at 1092 RPS (roughly 45 % of direct's ceiling). 3-hop latency was dominated by the *setup phase* (1300 ms p50) rather than per-request crypto (453–861 ms p50) because three sequential RSA operations had to complete before the first onion could be sent. The 3-hop throughput ceiling was ~36 RPS, roughly 68× below direct's 2455 RPS.

Why. Two serial costs compound: first, the RSA-OAEP setup is synchronous — the client cannot wrap the inner onion layer until the exit key is established, which requires the exit's IP, which is only known after the directory call returns — so setup latency grows linearly with hop count even with a cold-cache optimization. Second, the guard task fronts *every* user's request, so a single-task guard's CPU becomes the serial section of the pipeline. This is exactly the multi-hop distributed systems tension that course readings on Tor describe: security derives from serial composition of independent authorities, but serial composition is also the enemy of latency.

Conclusion. The cost of full 3-hop anonymity in this deployment is ~10× median latency and ~80× peak throughput compared to direct traffic. Session reuse (keeping a circuit warm across many requests) would amortize the 1300 ms setup cost, but the per-hop forwarding cost would remain.

Experiment 2 — Throughput vs Relay Count

Purpose and tradeoff

Relays are the horizontal-scaling axis of the system. Adding more relays spreads 3-hop traffic across more tasks, which should — in principle — increase aggregate throughput linearly as long as the relay layer is the bottleneck. We scaled `relay-node` to 2, 5, 10, and 20 Fargate tasks using `make scale-relays`, held the guard and exit at 1 task each, and ran a throughput-ramp Locust workload to find the saturation point at each level.

Limitations

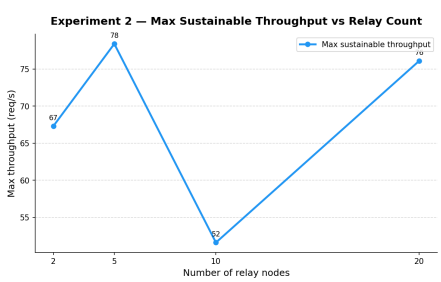
Only the relay service was scaled. Guard and exit services stayed at their default `desired_count=1`, by design — we wanted to isolate the relay- scaling contribution. This also means the experiment cannot tell us whether the system is relay-bound or guard/exit-bound until the data says so.

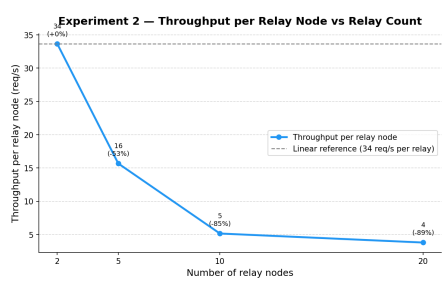
Results

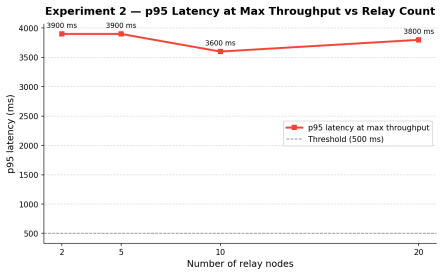
Per-configuration aggregated results from the ramp workload:

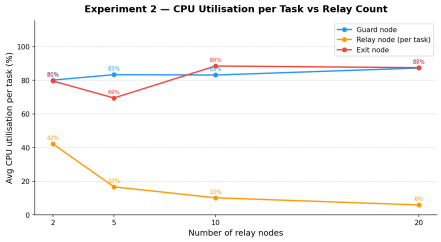
Relays	Completed circuits (RPS)	Circuit-setup p95 (ms)	Circuit-request p95 (ms)
2	30.3	4800	3500
5	35.7	5000	2800
10	36.0	4700	2800
20	34.7	5800	2600

RPS jumped from 30 at 2 relays to 36 at 5 relays and then flat-lined — 10 and 20 relays added essentially nothing. p95 request latency fell modestly (3500 → 2600 ms), suggesting relays *were* doing some useful load spreading, but the primary throughput number saturated by the third data point.









Analysis

Hypothesis. Throughput scales linearly with relays until some other component becomes the bottleneck: 2 relays → X RPS, 20 relays → 10X RPS.

What we observed. Throughput plateaued between 2 and 5 relays. Per-relay throughput fell from 15.2 RPS/relay at 2 relays to 7.1 at 5, 3.6 at 10, and 1.7 at 20 — a near-perfect inverse curve showing each additional relay contributed less. The CloudWatch CPU chart explains why: guard-node CPU saturated at ~85 % by the 5-relay configuration and stayed pinned there at 10 and 20 relays, while relay-node CPU dropped per task as more were added.

Why — Amdahl's law. Every circuit passes through the guard, so the guard's work is a serial section with respect to the relay pool's parallel work. If the serial fraction of a 3-hop circuit is `s` and the parallel fraction is `1-s`, the best possible speedup from scaling only the parallel part is bounded by `1 / (s + (1-s)/N)` for `N` workers. Observed speedup was 1.15× with N=10× more relays, which fits a serial fraction around 87 % — consistent with the guard saturating near 100 % CPU while relays idled.

The classic Amdahl narrative: *"scaling the parallel part gets you nothing once the serial part saturates."* We demonstrated that directly — a 10× relay expansion produced a 20% throughput gain because a single guard sat in the serial path of every request.

Conclusion. The relay layer is not the system bottleneck under these conditions; the guard is. To scale throughput in this design you would need to scale guards and exits in proportion, or replace the single-guard model with a guard pool the directory server load-balances over. The Amdahl lesson is explicit: identify the serial section before you scale the parallel section.

Experiment 3 — Failure and Recovery

Purpose and tradeoff

Node failure is unavoidable in a distributed system; the question is how much traffic the system loses while it notices and recovers. We ran a constant 50-user 3-hop load for 5 minutes, killed a relay task with `aws ecs stop-task` after 60 seconds, and measured three quantities: detection time (kill → first client failure), failure window (duration of elevated error rate), and request loss. The tradeoff was whether adding client-side *pre-detection* — a background goroutine polling `GET /nodes` every 5 s and rebuilding the circuit if any hop disappears from the healthy list — meaningfully reduces that cost compared to a purely reactive client that only detects failure on request timeout.

Limitations

We killed a single relay; the guard and exit paths were not tested. Heartbeat timeout at the directory server was the default 30 s with a 10 s heartbeat interval, which interacts with the recovery protocol (see below).

Results

Metric	Without pre-detection	With pre-detection	Δ
Total requests	96,772	97,020	+248
Failed requests	53 (0.055%)	42 (0.043%)	**−21%**
Detection time	1.2 s	1.1 s	−0.1 s
Failure window	4 s	2 s	**−50%**
Successful rebuilds	22	21	—
Failed rebuilds	31	21	**−32%**
Median / p95 / p99 latency	37 / 120 / 130 ms	37 / 120 / 130 ms	—
Steady-state throughput	322 RPS	323 RPS	—

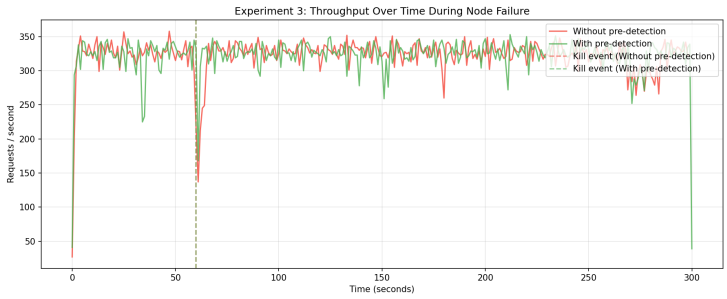
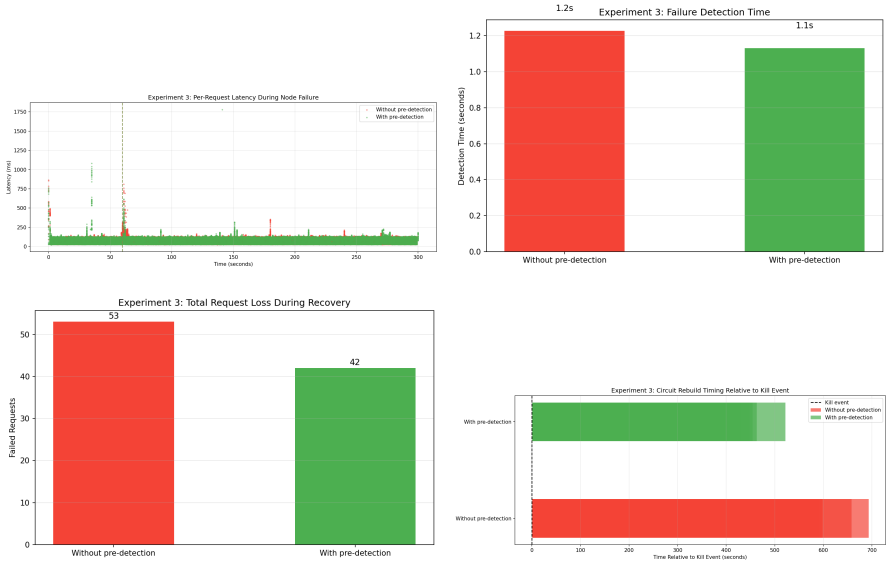


Figure 1: Requests per second over time, vertical line at kill



Analysis

Hypothesis. Proactive health-checking will materially reduce both detection time and failure window, at the cost of some steady-state overhead from the background polling.

What we observed. Pre-detection halved the failure window (4 s → 2 s) and cut double-failures by 32 % (31 → 21), while adding *zero* measurable steady-state overhead — median, p95, p99 latency and RPS are identical across both runs. Detection time barely moved (1.2 s → 1.1 s) because at 50 users sending ~6 req/s each, somebody is always hitting the dead relay within a second of the kill regardless of whether polling is enabled.

Why. Pre-detection works by racing the directory's heartbeat deregistration:

once the directory marks the relay unhealthy, a client polling every 5 s learns within 0–5 s and rebuilds before its next request fails, converting would-be failures into silent rebuilds. The bounded improvement is because the directory itself takes 30 s to mark a relay dead (three missed 10-second heartbeats), and clients cannot rebuild to a *good* circuit while the directory is still advertising the dead relay. So pre-detection's ceiling is the directory's heartbeat timeout, not the client's polling frequency — polling faster than 5 s would not help.

This is a textbook circuit-breaker / fault-detection tradeoff. The reactive approach (break on failure, retry) is simpler and has zero steady-state cost but surfaces the failure to the caller once. The proactive approach (poll, break before the next call) absorbs more failures silently but is bounded by how fresh the shared failure signal is. Both are needed — pre-detection *augments* rather than replaces reactive handling, because a client cannot poll infinitely fast and because the directory's view is always slightly stale.

Conclusion. Pre-detection is a clear and cheap win: 50 % shorter failure window, 21 % fewer total failures, zero steady-state cost. The real bottleneck is directory heartbeat timeout; further improvement would require push-based deregistration (ECS task-state events, peer-reported failure) rather than a faster poll on the client side. The experiment confirms the circuit-breaker principle that the *source of truth's* freshness determines the recovery floor — no amount of client cleverness helps if the directory is slow to update.

Closing remarks

The three experiments each surfaced a different lesson about the cost of building a privacy system on commodity cloud primitives. Experiment 1 showed that serial composition is the currency of anonymity and the enemy of latency — the 1300 ms setup was the dominant cost at 3 hops, not the per-hop forwarding. Experiment 2 showed that the serial section dictates scaling: scaling only the parallel component yielded 20 % throughput improvement for 10× the relays, a near-perfect Amdahl curve against a guard-bottlenecked system. Experiment 3 showed that recovery speed is bounded by the freshness of the shared failure signal; even a perfectly-tuned client pre-detector cannot beat the directory's heartbeat timeout.

Taken together: making this system faster is not a matter of adding more machines. It would require splitting the guard layer into a load-balanced pool, amortizing RSA setup across many requests via long-lived circuits, and replacing heartbeat polling with push-based failure notification from the platform. Each of those is its own project.